

On the use of passing network indicators to predict football outcomes[☆]

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ABSTRACT

Summary statistics for football matches, such as ball possession and percentage of completed passes, are not always satisfyingly informative about team strategies seen on the pitch. Passing networks and their structural features can be used to evaluate the style of play in terms of passing behavior, analyzing and quantifying interactions among players. The aim of the present paper is to show how information retrieved from passing networks can have a significant impact on the match outcome. At a descriptive level, we provide useful graphic visualizations to compare teams and their individual level of connection. Therefore, we directly compute and discuss network properties, such as centralization, clustering and cliques, from a football perspective. Then, we model the probability of winning the game through four competitive machine learning models including network-based indicators as explanatory variables with a set of in-field variables. The real dataset for application includes 96 matches in the Group Stage of the 2016–2017 UEFA Champions League, involving the 32 best European teams. This approach shows that some network-based variables, such as diameter and betweenness centralization, can be related to the level of offensive actions and finalizations for a team. Furthermore, we show that such variables help improve all considered models in terms of explanatory power, compared to those presenting only in-field regressors. Among the presented models, binomial logistic regression shows the best results according to a set of performance indicators.

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1. Introduction

In previous decades, team sport decisions often were subjective, trusting in the intuition and personal experience of decision makers, especially when they tried to associate team strategies and style of play with sport outcomes in terms of matches' results. Since 1992, when the American Statistical Association (ASA) created a section dedicated to sport statistics, the use of statistical techniques in sports have been promoted. Data-driven decision-making came to media attention in 2002, when Oakland Athletics General Manager Billy Beane adopted quantitative methods to assemble a competitive team from a cost-effectiveness perspective.

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Today, the impact of big data analytics extended the methodology to a broad range of sectors, becoming popular in other team sports (e.g., basketball, cricket, hockey) and, recently, in football as described in [1,2]. In addition to the individual skills of players, tactics and team strategies are key elements for success in football, and appropriate methodologies for dealing with these elements are still under debate.

Generally, football outcomes at the team level can be expressed in terms of results, goals scored or differences in goals or probability of winning [3,4]. Thus, statistical methods can be used to detect determinants and predict football outcomes. Some approaches consider ball possession the outcome variable of interest [5], while other works apply multivariate analysis, such as principal components analysis (PCA) and clustering, to summarize the amount of game-related statistics to distinguish the best clubs from each other [6] or to examine players' market value to determine how much either a team's reputation or individual skills affect the market values [7].

Another approach consists of the use of machine learning tools to predict match outcomes, trying to take into account several features that may influence the results of matches [see e.g. 8,9, and references therein]. From a different point of view, [10] test for the presence of Granger causality of European rivalries from

a time-series perspective. To predict match outcomes, it could be worth to use previous knowledge (often based on subjective judgments) obtained from experts in the field. In this framework, Bayesian networks have been applied in recent years [see, e.g., 11–13]; such methods provide promising results in terms of accuracy.

However, not all methods consider the strength of the relationship between players, the type of interactions among them in both at the micro-level (players) and macro-level (team) and the global team organization. In fact, passes and the related network are undoubtedly representative of game style, even compared to goals, shots and other summary statistics, representing more than 80% of the events in football [14]. Furthermore, network metrics, such as centrality measures and structural indices, have become more popular recently in the football area, as highlighted in [15] and [16]. Recent works also analyze a large available dataset, predicting football outcomes using players' abilities [17], including the evaluation on their passing skills.

The main innovative contribution of this paper is to discuss whether if and how passing network properties and performance indicators derived from network summary measures are crucial for the match outcome. Such measures may be able to improve statistical models where the football outcome is defined as the probability of winning the game.

Analyzing passing networks has some advantages: It provides an easy way to detect patterns or strong and weak ties among players and their positions in the lineup, can provide useful evidence of players' skills, teams tactics and connections between positions. Although information retrieved from this structure is not directly able to provide spatial information about the pitch, passing networks can be plotted clearly, helping to understand team cohesion from the micro- and macro-perspective. Finally, data are generally open access at the individual level for the main international and European tournaments.

Specifically, we aim to compare different statistical models to predict the probability of winning a match by using (to the best of our knowledge, for the first time) performance indicators based on passing networks along with the usual match-specific covariates, such as the number of fouls committed, teams' distance running performance, or attempted shots, among others.

We apply binomial logistic regression (BLR) [18] to model the probability of winning the game, showing how information extracted from passing networks can be used to improve model performances, evaluated in terms of sensitivity, specificity, accuracy and the area under the curve (AUC). Furthermore, to assess our approach, we compare BLR with other machine learning (nonparametric) methods used in the current literature to predict the probability of winning the match. Moreover, we check that fitted models give reliable predictions through the use of cross-validation techniques.

The rest of the paper is organized as follows: In Section 2, we present how to visualize passing networks emphasizing ties among players and their positions (micro-analysis); then we illustrate network summary measures along with their interpretation in terms of team sports (macro- and meso-analysis). In Section 3, we discuss how to model the probability of winning the game through BLR and three alternative nonparametric models. In Section 4, we discuss the main results of the proposed models applied to a real dataset for the European Champions League (Season 2016–2017), and finally, in Section 5, we summarize some concluding remarks.

2. Team passing networks

Network analysis deals with relational data, emphasizing investigation of the structure generated between units, driven by

Table 1

An example of an adjacency matrix in football.

	p_1	p_2	p_3	p_4	p_5	p_6	p_7	p_8	p_9	p_{10}	p_{11}
p_1		0	4	0	0	4	4	2	0	0	0
p_2	0		0	0	0	0	5	0	0	2	0
p_3	0	0		0	0	2	0	0	0	0	7
p_4	0	4	5		0	0	2	0	0	4	0
p_5	2	0	0	1		0	0	0	0	0	0
p_6	0	0	0	0	0		0	4	1	3	0
p_7	0	0	0	0	0	0		0	0	0	1
p_8	0	0	0	0	0	0	0		0	1	4
p_9	0	0	0	0	2	0	0	2		0	3
p_{10}	0	0	6	0	0	0	3	2	0		0
p_{11}	0	0	0	0	4	0	0	0	0	0	

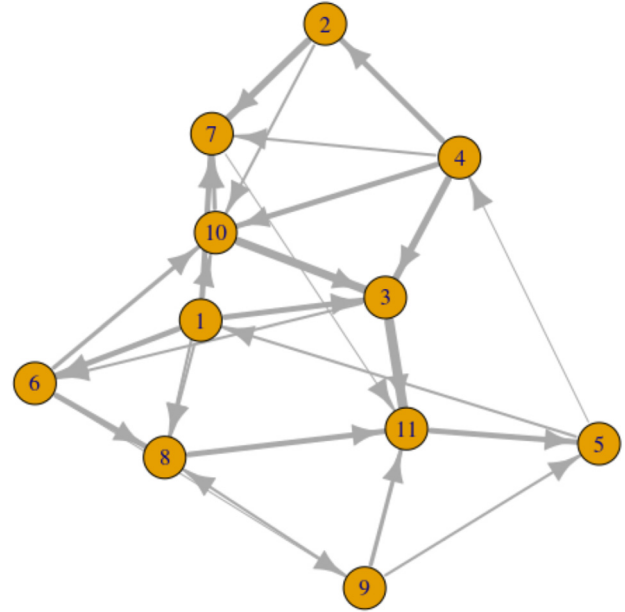


Fig. 1. Graph of the passing network in Table 1.

the quantity and quality of the ties occurring among them. Network theory includes all possible methods for analyzing data which present interactions between a set of units (agents) to investigate patterns and community structures [19].

Networks can be modeled in terms of graph theory. In this framework, a network G is defined as the ordered triple $(\mathcal{V}, \mathcal{A}, \mathcal{W})$ consisting of a set of nodes $v \in \mathcal{V}$, a set of arcs $a \in \mathcal{A} \subseteq \mathcal{V} \times \mathcal{V}$ and a set of weights $\omega \in \mathcal{W}$. We assume that the vertices $\mathcal{V}$ and arcs $\mathcal{A}$ are finite, as well as $|\mathcal{V}| = n$. An ordered pair of vertices of $\mathcal{V}$ denotes an arc through a function $\psi: \psi(v_i, v_j) = a_{ij} \in \mathcal{A}$, mapping the one-directional tie from vertex v_i to v_j , with $i, j = 1, \dots, n$. The mapping $\omega: \mathcal{A} \rightarrow \mathbb{R}$ defines the weight related to each arc. A network can be expressed in the form of an adjacency matrix $P = (p_{ij})$, with $p_{ij} = \omega(a_{ij})$ if $\exists a_{ij} \in \mathcal{A}$, and $p_{ij} = 0$ otherwise. In our case, the nodes are the 11 first team players, linked by a directed relationship, i.e., a passing relationship, and weights are the number of passes that occurred during the match, denoting the strength of the connection. A first descriptive comparison between footballers can be provided in terms of a graph plot, and, additionally, it is possible to consider the vertices' positions to eyeball the cohesion of the players and structural strategies.

An example of an adjacency matrix of team passing network distribution is presented in Table 1 which contains a one-mode squared and asymmetric matrix. Fig. 1 represents the directed and weighted graph obtained through the adjacency matrix in

Table 1. Eleven agents (or players) are connected by the arrows, and the strength of the relationship (i.e., the number of passes) is expressed through the width of the arcs. Moreover, the values on the main diagonal are constrained to be null, because we assume that a player cannot pass the ball to himself.² This framework allows us to make comparisons at the micro-level involving players, but some information can be also extracted in a general context, to evaluate the global performance of a team.

To provide a more readable passing graph, we normalize the weights with respect to all the passes that occurred among players during the entire match:

$$\tilde{\omega}_{ij} = \frac{p_{ij}}{\sum_{i=1}^n \sum_{j=1}^n p_{ij}}, \quad (1)$$

and then we emphasize only the ties whose weights are greater than the theoretical value related to a fully connected directed network: $1/e_{max}$, where $e_{max} = n(n - 1)$.

2.1. Performance indicators derived from passing networks

Topological properties of the network can help to identify relevant structures and hidden features underlying passes in football (or even in other team sports), and can be profitably used to obtain performance measures at the team and the player level. The literature includes numerous structural indices able to describe networks [see 20–22, among others]; some can be applied to team sport passing networks and can be interpreted in terms of performance indicators [see e.g. 23]. For our purposes, we focus on a restricted number of indices, called network summary measures, which are able to capture the complexity of the network topology and have a meaningful interpretation for football and other team sports. Note that some of the used indices are designed for undirected and binary networks.

Network summary measures can be divided into three main categories: Group-based (or macro-) indices describe the overall characteristics of a network, while micro-indices focus on the individual nodes. Measures that combine information from the micro- and macro-levels are usually denoted as meso-level measures.

Macro-level

One of the simplest team-level summary indices is the *total number of links*, i.e., the number of links occurring in the network, corresponding to the absolute number of total passes conducted between teammates during the match. A higher value of this index reveals strong cooperation between team players, who successfully interact with each other. However, this measure reflects only partial information that can be extracted from the whole network. A review of the most relevant network indices can be found in [19].

Network diameter is expressed as the *geodesic* distance between the most distant nodes of a graph, representing the extent of the graph and the topological length between the two most distant nodes, without taking into account the link weights. A team with a high network diameter reflects the team's ability to generate as many direct connections as possible, even in terms of passes.

Reciprocity index is computed as the proportion of mutual connections in a directed graph, i.e., the probability that the opposite counterpart of a directed edge is included in the graph. In practice, this index measures the ability of two players to have mutual connections. A team expressing many null or reciprocated

ties may be different from one that has asymmetric connections (which might be a hierarchy).

The *assortativity* coefficient measures the level of homophily for a graph; this coefficient, based on the degree of vertices, is normalized and takes values in the interval $[-1, 1]$. High assortativity means that players who have an equal or similar number of passes are often connected. Otherwise, low-degree players are more likely be connected to teammates with a high degree of passing [for further details on the assortativity coefficient, see 24].

Micro-level

Several individual indices involve subgroups of players considering complete subgraphs within a network, also denoted as *cliques*. Cliques can be helpful to understand the structure of interactions between teammates and the regularity of such connections. Therefore, two useful indices can be computed: the size of the *largest clique*, representing the number of vertices which contains the largest clique, and the number of many times the *maximal clique* occurs, denoted as the *countmax cliques*. In this case, it is possible to identify the size and regularity of the denser passing structures occurring in a match. However, when measuring cliques, directional properties may be lost.

The generalized *clustering coefficient* [25], based on the transitivity property, is a measure used to detect the fraction of closest triplets in directed networks. The coefficient is also a normalized index and takes a value within the interval $[0, 1]$. When the value is closer to one, the passing triplets between teammates are dense. This index can be used to identify patterns of interactions between teammates based on their so-called dyadic relationship.

One of the most relevant roles of network analysis is the identifying the most influential nodes. A set of useful measures in this context is represented by the class of *centrality* measures [26]. There are different ways to interpret which are influential nodes in a network, for this reason the concept of centrality is not uniquely defined. For this purpose, various measures have been proposed, focusing on different aspects of the network topology. Among them, degree and betweenness centrality are widely used in academic research. *Degree* centrality represents the number of edges incident upon a given node and can be easily computed as the marginals of the adjacency matrix. This measure belongs to a broader class of radial measures, meaning that the score is computed starting from a given node. *Betweenness* centrality [26, 27], considered one of the most reliable measures, is based on the number of paths that pass through a given node. The various centrality measures in the literature include the *eigenvector* centrality [28], based on the singular value decomposition of the adjacency matrix. In conclusion, the high score of these indices are associated with the most central vertices of the network.

Meso-level

A third approach for extracting information from a network is given by summarizing player-level indices to retrieve proper network summary measures. For this purpose, the first index considered is the *diversity* of a node, i.e., Shannon's entropy of the weights regarding the incident links of a node. The diversity index can be useful for understanding players' similarities; in practice, we compute the median value of the entropy among the nodes.

Therefore, the *average neighborhood* is the mean of the nearest neighbor degree of each vertex. This index represents the average of the degrees of the partners of *i-th* node; i.e., it provides a measure of the connectivity of the neighbors of a certain player. A global index was obtained by computing the median of these values.

² This type of connection is usually denoted as a "loop".

The definition of centrality on the node level introduced previously can be extended to the whole graph, often denoted as graph *centralization*. This class of indices evaluates whether the entire network is able to express a star-like topology, which means that the nodes show, on average, the same level of connectivity. If this index is close to zero, the interactions between players are homogeneous, while centralization near to one suggests a star-like topology. To summarize, when the centralization is close to one, teammates tend to pass often to the same player. The centralization index strongly depends on which node-level centrality measure is chosen to compare the relevance of different nodes in a graph. However, although centrality measures identify the most important node within a graph, centralization refers to structural relationships among all vertices.

The simplest measure to be computed is the *degree* centralization index, which relies on the number of edges incident upon a given node. This index can be easily computed as rows' or columns' marginal of the adjacency matrix. Another class of measures is based on the number of paths occurring through a given node: The best-known is the *betweenness* centralization. From a football perspective, players with higher betweenness scores are more influential in terms of passes among other players, acting as mediators or bridges, playing an important role in passing the ball to the other players. For instance, central midfielders and defenders are expected to have the highest betweenness scores.

A last set of measures occurring in directed networks consider two types of vertices called *hubs* and *authorities*. To summarize, a vertex can be a good hub if it points to many good authorities or a good authority if it is pointed by many good hubs. The *pagerank*, introduced in [29] and successfully applied to rank web pages, is one of the most popular meso-level summary measures. It can help uncover influential or important nodes whose reach extends beyond their direct connections, taking directions and weights into account. In the team passing network structure, this index is capable of detecting central players in terms of passes: A player is considered "important" (thus, he will have a high pagerank score) if he is linked to by other central players or if he is highly connected. In conclusion, these measures are summarized up to the team level by taking the 75th percentile of corresponding node-level values.

3. Modeling football outcomes

To understand the relationship between the passing network structure, network measures and football outcomes, we use quantitative methods that consider different probability assumptions. Statistical models have been recently developed to make inferences about football outcomes considering several dependent variables, such as the probability of winning a game (but also the exact result), goals scored, the difference between goals and, sometimes, even ball possession. The current literature is mainly focused on the prediction of goals scored between two competing teams. Goals scored can be modeled separately [see e.g. 30,31], or the differences between the number of goals scored can be considered [as in 32], which often involves Bayesian techniques. Other approaches consider football matches as pairwise comparisons; these approaches are focused on the estimation of global ability (or strength) to rank teams [33].

Although existing methods are mainly focused on the prediction of football outcomes, often with the additional purpose of quantifying and estimating an economic return for betting strategies, our approach consists of quantifying how network variables (expressed in terms of summary measures) can be used as explanatory variables for the probability of winning the game. In our type of post-hoc analysis, we include team-level control variables, as suggested in the current literature [34], considering

popular match statistics, usually freely available at the end of each match.

We want to test whether there are network indicators that can provide useful information about passing behavior to assess the strength of a team, directly influencing its probability of winning the game, and that can improve the capability predicting match outcomes. We focus on the conventional BLR with k covariates, which is one of the most applied generalized linear models [see also 18, for further details on modeling categorical outcomes]. This method allows us to (i) investigate the determinants of winning the match, (ii) quantify the effects of statistically significant variables, which may include network structural properties, and (iii) assess the goodness of fit in terms of correct prediction rates.

The conventional BLR can be summarized with the following expression:

$$\text{logit}[\pi(\mathbf{x})] = \log \left[\frac{\pi(\mathbf{x})}{1 - \pi(\mathbf{x})} \right] = \beta_0 + \beta_1 x_{1i} + \dots + \beta_k x_{ki}, \quad (2)$$

where $\pi(\mathbf{x}) = P(y = 1|\mathbf{x})$ is the probability of winning the game given model covariates, and the $k \times 1$ vector of parameters is $\beta = \{\beta_0, \beta_1, \dots, \beta_k\}$. The quantities of interest are the exponential of the structural parameter which can be directly interpreted in terms of the odds ratio (OR) for the probability of winning. Estimates of parameters are obtained by applying the R function `glm`, through maximum likelihood estimation.

To assess the performances of this model, we compare our results with three different nonparametric classifiers applied in the literature [see e.g. 35].

Artificial Neural Network (ANN) represents a broad class of nonlinear models inspired by the structure of biological neural networks, introduced by [36]. Since then, this class of models developed impressively during the last few decades. In our work, we adopt the most popular architecture in the majority of current applications, denoted as the "backpropagation feedforward" ANN [37].

Naive Bayes (NB) is one of the simplest and most-studied probabilistic classifiers in the literature [cfr. 38,39] and computes the probabilities according to Bayes' rule. The algorithm assigns the most likely class to a given example described by its feature vector by using as discriminant functions the class posterior probabilities, given the feature vector itself.

Random Forest (RF) introduced in [40], consists of a tree ensemble learning model based on the generation of a series of uncorrelated trees combined with randomized node optimization and bootstrap aggregating to improve the stability and accuracy of the model. In this framework, decision trees are generated by using well-known splitting criteria, such as information gain or Gini impurity [41].

The models above are tuned to be suitable for the experimental dataset to ensure a fair comparison. The evaluation of BLR and nonparametric models is conducted in terms of predictive ability. For this purpose, we compare predictions implied by the models against actual values using three well-known measures. These indicators represent a synthesis of the so-called predictions/realizations table, also denoted as the "confusion matrix". Table 2 shows a generic 2×2 confusion matrix, presenting actual values in raw and predictions (fitted values) in columns, which summarizes the classification problem and allows to compute three indicators:

Sensitivity, the relative frequency of games correctly classified as a loss or draw:

$$\text{Sen} = \frac{n_{00}}{n_{.0}},$$

Table 2
Prediction/realization table (confusion matrix).

True \ Predict	$\hat{y} = 0$	$\hat{y} = 1$	Tot.
$y = 0$	n_{00}	n_{01}	$n_{0.}$
$y = 1$	n_{10}	n_{11}	$n_{1.}$
Tot.	$n_{.0}$	$n_{.1}$	n

Specificity, the relative frequency of games correctly classified as a win:

$$Spe = \frac{n_{11}}{n_{1.}},$$

Accuracy, the overall rate of correct classification for the whole model:

$$Acc = \frac{n_{00} + n_{11}}{n}.$$

The AUC represents another useful indicator. Given a receiver operating characteristic (ROC) curve [see also 42] showing the performance of a classification model applying different thresholds, the AUC measures the entire two-dimensional area underneath the ROC curve. This measure tells us how much a model is capable of distinguishing between classes. Formally, the AUC is defined as:

$$AUC = \int_0^1 ROC(u)du.$$

As use of the whole dataset can lead to the overfitting issue in terms of predictive evaluation, especially in the nonparametric models (such as an ANN and RF), it is often suggested to split the sample into two parts. This procedure consists of estimating the model through with the first subsample, denoted as the “training” set, and then assessing the predictive power of the model using the second subsample, the so-called “test” set. To avoid choosing a (possibly arbitrary) sample split criterion, we apply a robust procedure generating different splits. This method is a particular type of cross-validation (CV) technique [43] also denoted as Monte Carlo repeated random sub-sampling validation [cfr. 44], i.e., MCCV. We use these repeated and randomly chosen training and test sets to compute B times (where B is the number of Monte Carlo replications) the sensitivity, specificity, accuracy and AUC, reporting their averaged values, and then estimate their uncertainty in terms of the standard deviation among the B replications.

4. Real data application

We apply previously introduced methods to the passing distributions from the Group Stage of 32-team UEFA Champions League (UCL) in Season 2016–2017. Data are collected using freely available press kits through the official UEFA website (www.uefa.com). They include 96 matches and 192 passing networks for 32 European teams and present the same structure, i.e., the adjacency matrix illustrated in Section 2 in Table 1 and Fig. 2.

For each team, we numerically compute all indices introduced in Section 2.1 for six matches from passing networks. Thus, descriptive statistics, such as the mean, standard deviation (S.D.), minimum (Min) and maximum (Max) are depicted in Table 3. In the results some indices are numbers, e.g. cliques or number of passes, while others are constrained to assume values between 0 and 1, such as centralization, hub and authority, or between -1 and 1 (e.g., assortativity coefficient). Descriptive analysis confirms the key role of the number of passes in identifying the strength of a team: in Table 4, we include only the five best and worst teams in the first phase of the tournament, highlighting the best and worst performances.

Table 3

Descriptive statistics of network indicators in 2016–2017 UCL (Group stage).

	Mean	S.D.	Min	Max
Number of passes	386.25	136.93	121.00	876.00
Diameter	6.04	1.47	3.00	10.00
Largest clique	8.10	1.05	5.00	11.00
Countmax cliques	6.36	3.42	1.00	20.00
Reciprocity	0.75	0.09	0.42	0.94
Global clustering	0.82	0.06	0.59	0.93
Diversity	0.92	0.02	0.88	0.95
Centralization betweenness	0.04	0.02	0.01	0.15
Centralization closeness	0.17	0.05	0.08	0.30
Centralization degree	0.19	0.05	0.06	0.39
Centralization eigenvalue	0.20	0.06	0.08	0.39
Hub	0.72	0.11	0.40	0.95
Authority	0.77	0.11	0.45	0.96
Pagerank	0.11	0.01	0.10	0.14
Assortativity	−0.17	0.06	−0.31	0.03
Average neighborhood	17.20	2.13	12.40	25.09

Table 4

Total completed passes by a team in each match, 2016–2017 UCL (Group stage).

Team	I	II	III	IV	V	VI	Total
1 Bayern München	721	612	787	728	645	759	4252
2 Barcelona	772	615	383	521	649	939	3879
3 PSG	519	725	647	658	557	534	3640
4 Borussia Dortmund	580	603	451	519	737	533	3423
5 Juventus	448	744	492	515	467	618	3284
...	...	...	...	...	...	...	...
28 Legia Warsav	221	336	349	452	254	278	1890
29 CSKA Moskov	244	237	295	314	388	218	1696
30 Leicester	275	223	277	306	370	213	1664
31 Dynamo Zagreb	341	246	167	203	285	328	1570
32 Rostov	179	370	213	197	130	223	1312

Following the model setting introduced in Section 3, we include some control variables: The first is UEFA *Ranking* of each team in the season, which was categorized in four classes. Then we consider the *Home/Away* dummy variable to take into account the home effect. We also collect the following *in-field* variables: *attempts* (i.e., the number of attempted shots in a match per team), *offsides* (defined as the number of offsides committed), and *fouls committed* (which is the raw number of fouls committed including hands). Finally, we include ball possession (in percentage) and the *distance* (in kilometers) totaled for all players on a team to have a synthetic measure of athletic skills.

The total number of passes reflect in this way the competitiveness of a team; all of the five top-ranked teams in terms of passes advanced to the knockout phase, while a four-fifths of the worst five teams failed to pass the group stage. An exception is Leicester City: The team passed through the round of 16 and got to round of 8 although it showed a very poor performance in terms of completed passes and accuracy.

4.1. Empirical illustration of team passing networks

We present an empirical illustration of passing networks (micro-analysis) through two opposite examples, using the empirical visualization tools explained in Section 2. The main results of the analysis are developed through R Software using the packages *igraph* and *sna*.

The first example regards the passing network of Barcelona versus Borussia Mönchengladbach; in this match, “Blaugrana” completed 939 passes, with approximately 92% accuracy. The graph in Fig. 2 clearly shows the well-known Barcelona “tiki-taka” style of play based on ball possession and numerous short passes. We point out important ties using the empirical threshold procedure on weights proposed in Eq. (1); most relevant arcs are represented by solid lines, while dashed lines represent less

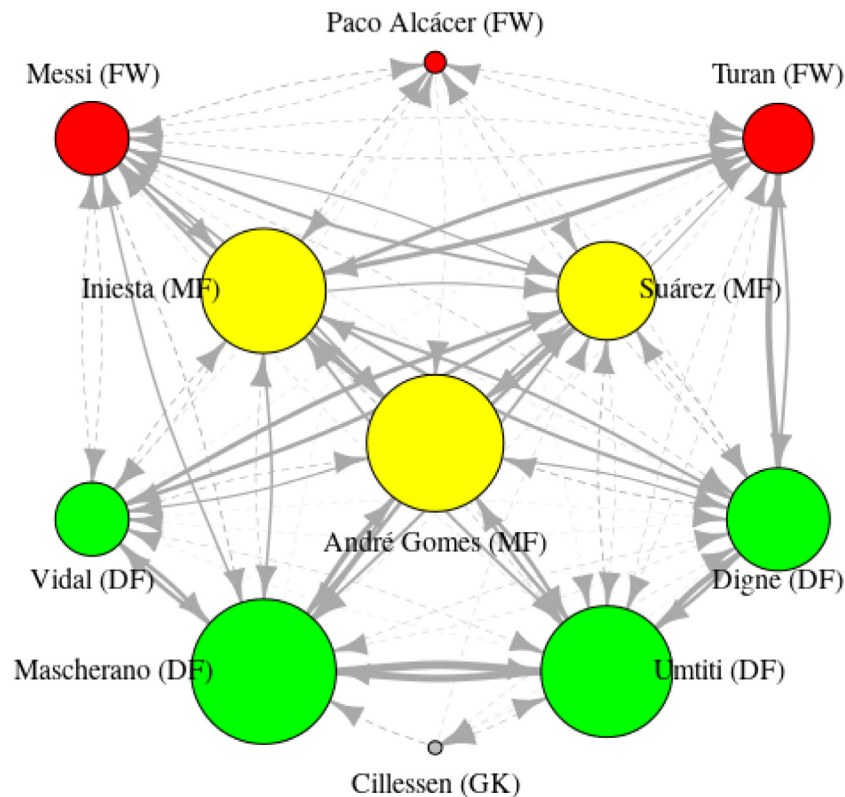


Fig. 2. Passing network of the Barcelona match: *Barcelona–Borussia Mönchengladbach 4-0*.

important links. To show the importance of players in a network in terms of out-degree centrality, we also set the size of all vertices depending on the number of completed passes to see which players are more involved in cooperation processes, e.g., comparing center defenders to midfielders.

Analyzing the out-degree centrality we observed 5 large-sized players, including the three midfielders, and the role of center defenders in starting the offensive actions. We also find a stronger relationship between Lionel Messi, who is one of the most skillful players, and midfielders Andrés Iniesta and Denis Suarez rather than the drawback player Arturo Vidal. We also clearly notice that right-wing Arda Turan mostly cooperates with his right back Lucas Digne. Furthermore, the graph highlights how center forward Paco Alcacer acts as a false nine³ rather than a striker,⁴ presenting low out-degree centrality with respect to other Barcelona players. It is possible to see that all his ties are weaker considering the empirical threshold, meaning that he exchanges the ball only with his closest players. Other similar considerations help to identify typical “Blaugrana” styles of play, the direct expression of modern Spanish football philosophy.

Fig. 3 illustrates Rostov’s passing network in the match against Bayern München, representing a very poor performance in terms of completed passes (297). The goalkeeper plays a central role in ball possession, preferring to jump the defender, tending to pass directly to the right midfielders. The outdegrees of central forwards are comparable with respect to the other players, while the central midfielder Alexandru Gatskan gives and receives many passes from all the positions. He could be considered the lighthouse of Rostov: Center defenders seem to have a less important role in possessing the ball, especially compared Barcelona’s

previous counterparts (Mascherano and Umtiti). To summarize, Rostov exhibits a totally different style of play due to tactical and technical reasons.

Even if passing networks may be very informative about football strategies, other features such as the degree of accuracy and the initial formations, e.g., 4–3–3 vs. 5–3–2, play a central role in ball possession. Shots, aggressive attitudes and the amount of running can be crucial in a high-quality teams’ tactics and generally, increasing the probability of winning the game. To clarify the relevance of network-related indices as explanatory variables of football outcomes, we focus on statistical modeling that has network summary measures as covariates.

4.2. Statistical models

To predict the probability of winning through network summary measures, in-field and control variables, we estimate a set of statistical models using previously discussed real data (Group Stage of 2016–2017 UCL). The first model includes all of these indicators in a (full) preliminary logistic model, and then we use a well-known automatic procedure, denoted as *stepwise* selection with bidirectional elimination, to (a) identify the most relevant set of variables in terms of (global) statistical significance and (b) avoid the possible issue of multicollinearity. The procedure is based on the evaluation of the Akaike information criterion (AIC). Thus, we compare this model with three nonparametric models (NB, ANN and RF) in terms of explanatory power (where possible) and predictive abilities.

In Table 7 is summarized the input scheme for different statistical models and classifiers explained in Section 3, divided into control variables and network summary measures. Each model was compared with its version that does not include the in-field variables as input (+N and -N in the table); while the symbol “√” indicates when a variable, depicted in the rows, is included in the statistical model. In particular, the BLR model refers to the

³ A false nine is an unconventional lone striker, characterized by good vision to play and short passing abilities to link up with the midfield.

⁴ A typical striker is an agile and fast forward player with good ball control and dribbling abilities.

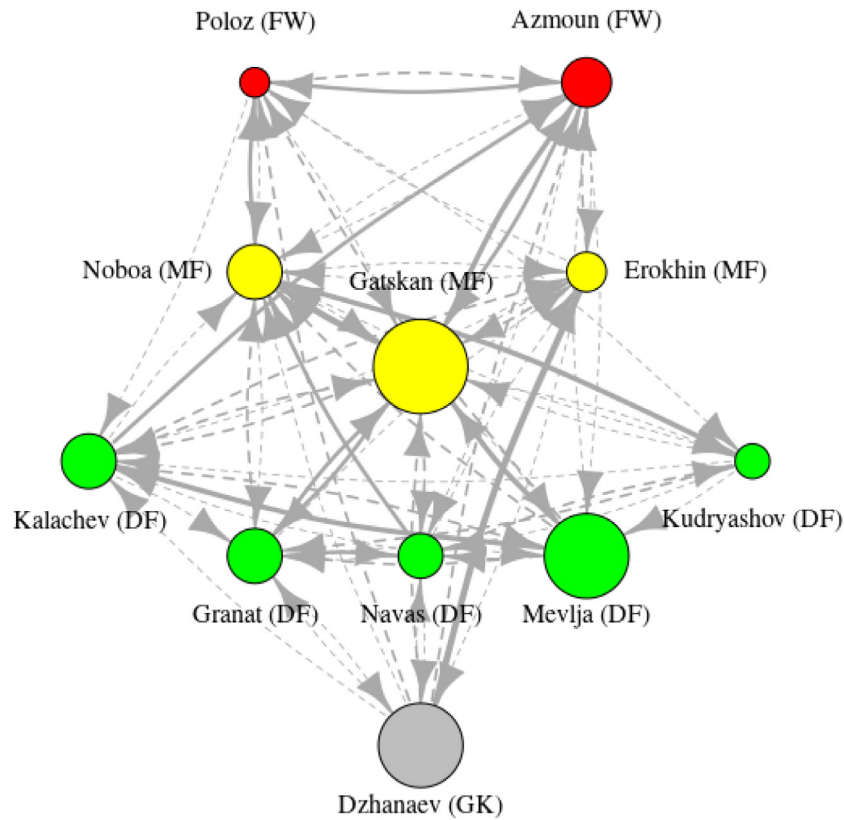


Fig. 3. Passing network of the Rostov match: FC Rostov–Bayern München 3–2.

model obtained after the stepwise selection algorithm, and for the sake of comparison, we also use the same set of variables for the NB algorithm (denoted as NB–RED) even if “frequentist” model selection is not a proper tool in Bayesian inference. For this reason, we also consider an NB complete model, involving all available variables (in-field, control and network related), as well as ANN and RF models.

The main output for the BLR model can be viewed in Table 5, it includes the estimates for the parameters (obtained through maximum likelihood estimators), their ORs computed as $\exp(\beta_j)$, standard errors, the values of z tests for the null hypothesis $H_0 : \beta_j = 0$ and p values related to asymptotic z statistics. The selected variables are the ranking (in four categories), number of attempted shots, fouls committed, number of passes and three of our network indicators (i.e., the diameter, the average neighborhood and the centralization betweenness index). All of these variables are statistically significant considering a type I error of $\alpha = 0.05$ except for average neighborhood (p value < 0.1) and fouls (p value ≈ 0.1). Lower ranking, fouls committed, diameter and average neighborhood have a negative impact on the probability of winning the game; for example, this probability increases about 9.6% for each attempt. Moreover, attempts, passes and betweenness centralization share a positive sign on the logit of wins. In particular, considering the ORs, for each additional point of centralization betweenness, the probability of winning the game increases about 35%, *ceteris paribus*.

To assess the predictive power of the models, we compare the BLR against all the proposed models from Section 3. Prediction indicators, such as the AUC, Sensitivity, Specificity and Accuracy are computed $B = 1000$ times by using a randomly chosen test set of size equal to $n_{test} = n \times 0.25$. In Table 6, we report the averages and standard deviations computed among $B = 1000$ replications.

First, we immediately notice that the BLR model performs well for all indicators with respect to its counterpart without network summary measures (denoted with $-N$ in the table), especially for the AUC and in terms of uncertainty (lower standard deviation). In particular, the Specificity (i.e., the ability to predict the winner) increases from the 53% to 58% when summary network indicators are added in the BLR model. Second, BLR presents the best performance in terms of accuracy (76%) and AUC (81%) in common with the NB–RED. Moreover, RF and NB outperform BLR, respectively, in terms of sensitivity and specificity, even if RF presents more unbalanced results (e.g., the specificity is lower than 50%). NB with network indicators reveals the greatest ability to predict the winner of a game, expressed through a specificity of 67%. The BLR model involving network measures can help explain and predict the outcome, being able to quantify the impact of explanatory variables in terms of ORs. Furthermore, BLR presents a better or comparable performance in terms of predictive power with respect to other widely used techniques.

Finally, for the BLR, ANN and RF models, we compare the variable importance. Specifically, for BLR we consider the z values, while for the two nonparametric models we use model-specific ranking scores. In particular, for the ANN, the variables' importance ranking in Fig. 4 were computed using combinations of the absolute values of the weights [45]. This method, depicted in [46] and [47], is useful for detecting the most influential variables of an RF model measuring the differences in prediction accuracy before and after permuting each input variable. The main results are depicted in Table 6.

Considering the network variables, the number of passes and betweenness centralization are relevant for all three models. Diameter and average neighborhood share an explanatory power to model the probability of winning for BLR and the ANN, but in the RF model, these variables are located in the lower part of the ranking. Regarding in-field variables, the variable “number of

Table 5
Output of binomial logistic regression (BLR).

	Estimate	Odds ratio	Std. error	z value	p value
(Intercept)	0.770	2.160	2.006	0.38	0.701
Ranking in National Federation_2	-0.861	0.423	0.450	-1.91	0.056
Ranking in National Federation_3	-1.702	0.182	0.622	-2.74	0.006
Ranking in National Federation_4	-2.221	0.108	0.721	-3.08	0.002
Attempts	0.092	1.096	0.040	2.30	0.021
Fouls committed	-0.071	0.931	0.043	-1.64	0.101
Number of passes	0.009	1.009	0.003	3.31	0.001
Diameter	-0.397	0.672	0.186	-2.14	0.033
Average neighborhood	-0.182	0.834	0.105	-1.72	0.085
Betweenness centralization	0.306	1.358	0.100	3.06	0.002

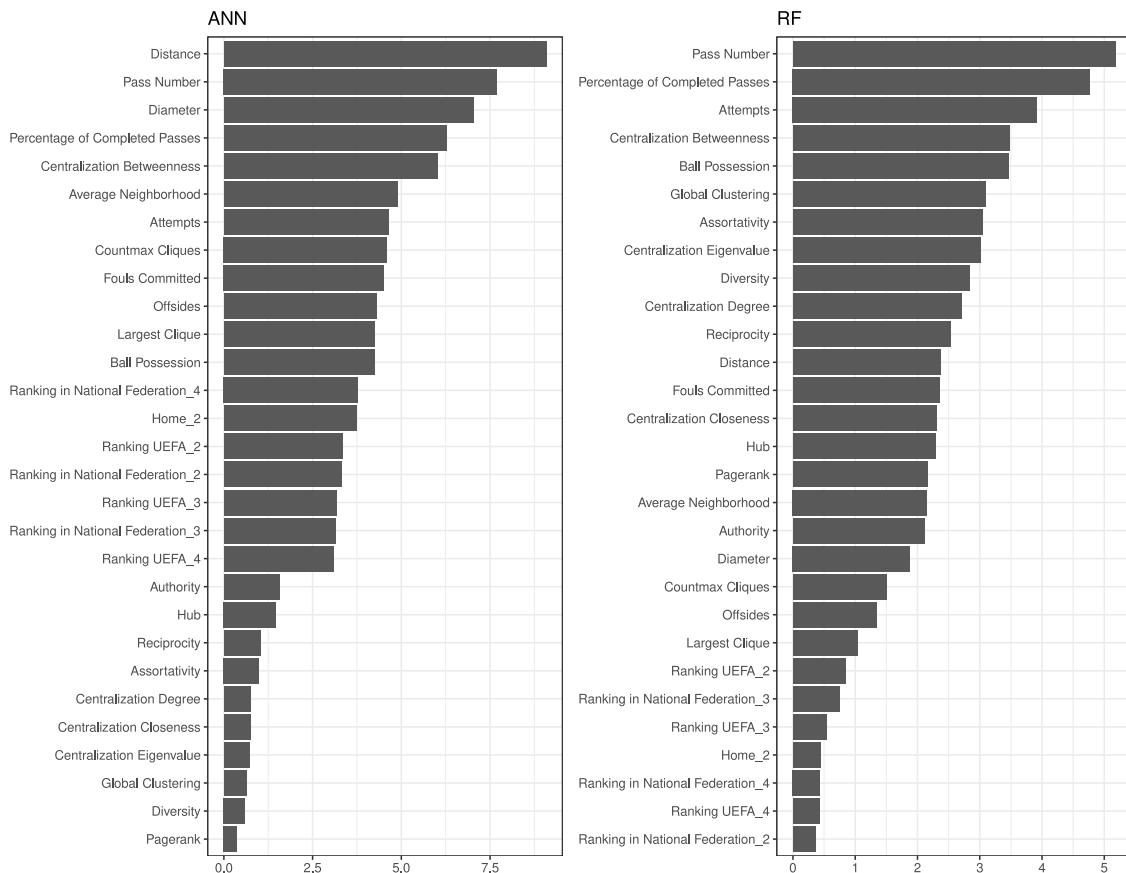


Fig. 4. Ranking of the variable importance of the ANN and RF models.

attempts" provides the largest contribution in all three models. In addition, the ranking in national federations has a very high z value for the BLR model, but the ranking it occupies the middle or lower part of the respective variable rankings for the two nonparametric models.

5. Conclusions

New technologies, such as wearable devices, multiple-camera player trackers and drone-based analysis of training sessions, are changing the way to retrieve data and provide new opportunities in football tactics. Starting from this point of view, we compute a set of structural and descriptive indicators (such as density, diameter and centralization measures) based on passing networks. We consider four classes of models, parametric and nonparametric, to verify whether and how network variables improve the conventional models that include only match statistics.

To the best of our knowledge, this is the first work that considers passing network indicators to model the probability

of winning the game and even comparing different statistical models. The main contribution lies in showing that a limited number of network variables, namely, related to the topology and structure of the global passing network, are relevant for all the considered models and yield some improvements in model performance. In particular, we discuss the density, the diameter, the betweenness centralization and the average neighborhood. These variables essentially measure the construction of offensive actions and finalizations. With respect to the analyzed models, given the performances in terms of accuracy and the AUC, BLR appears to be preferable even considering its parsimony and simplicity.

In this work, we showed how the structural passing network features could be informative for football teams' staff, managers and match analysts from the descriptive and predictive perspectives. To improve the probability of winning a match, a team can aim to increase the following scores: the length of interactions (diameter), the capability of playing a succession of passes at pace (betweenness centralization), high passing connection and

Table 6
Evaluation of statistical models.

	BLR		NB		NB-RED		ANN		RF	
	+N	-N	+N	-N	+N	-N	+N	-N	+N	-N
AUC	0.81 (0.06)	0.79 (0.06)	0.79 (0.07)	0.78 (0.06)	0.81 (0.06)	0.78 (0.06)	0.76 (0.11)	0.78 (0.14)	0.75 (0.07)	0.75 (0.06)
Sensitivity	0.87 (0.06)	0.85 (0.07)	0.80 (0.07)	0.79 (0.08)	0.84 (0.07)	0.85 (0.06)	0.83 (0.12)	0.84 (0.13)	0.90 (0.06)	0.87 (0.07)
Specificity	0.58 (0.13)	0.53 (0.11)	0.67 (0.11)	0.65 (0.10)	0.57 (0.12)	0.56 (0.10)	0.42 (0.24)	0.37 (0.25)	0.44 (0.12)	0.44 (0.15)
Accuracy	0.76 (0.06)	0.74 (0.05)	0.75 (0.06)	0.74 (0.06)	0.75 (0.06)	0.75 (0.05)	0.69 (0.07)	0.67 (0.07)	0.74 (0.06)	0.72 (0.06)

Notes: The table shows a summary of the main models' performance measures (along with their respective standard deviations in brackets), computed by averaging results obtained through the MCCV cross-validation procedure. Each model is compared with its respective version without network indices as the input (+N and -N). BLR represents the reduced logistic regression model, NB stands for the Naive Bayes approach computed on the full input dataset, while NB-RED stands for the Naive Bayes model that has the same set of regressors included in the logistic model; ANN gives the composition of the proposed artificial neural network input setting, and RF that of the random forest.

Table 7
Input variables setting scheme used in each model.

	Label	Description	BLR		NB-RED		NB		ANN		RF	
			+N	-N	+N	-N	+N	-N	+N	-N	+N	-N
In-Field Variables	X ₁	UEFA Ranking					✓	✓	✓	✓	✓	✓
	X ₂	Ranking in National Federation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	X ₃	Home					✓	✓	✓	✓	✓	✓
	X ₄	Attempts	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	X ₅	Offsides					✓	✓	✓	✓	✓	✓
	X ₆	Fouls Committed	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	X ₇	Ball Possession					✓	✓	✓	✓	✓	✓
	X ₈	Distance					✓	✓	✓	✓	✓	✓
Network Summary Measures	X ₉	Number of Passes	✓		✓		✓		✓		✓	
	X ₁₀	Diameter	✓		✓		✓		✓		✓	
	X ₁₁	Largest Clique					✓		✓		✓	
	X ₁₂	Countmax Cliques					✓		✓		✓	
	X ₁₃	Reciprocity					✓		✓		✓	
	X ₁₄	Global Clustering					✓		✓		✓	
	X ₁₅	Percentage of Successful Passes					✓		✓		✓	
	X ₁₆	Diversity					✓		✓		✓	
	X ₁₇	Hubs					✓		✓		✓	
	X ₁₈	Authorities					✓		✓		✓	
	X ₁₉	Pagerank					✓		✓		✓	
	X ₂₀	Assortativity					✓		✓		✓	
	X ₂₁	Average Neighborhood	✓		✓		✓		✓		✓	
	X ₂₂	Betweenness Centralization	✓		✓		✓		✓		✓	
	X ₂₃	Closeness Centralization					✓		✓		✓	
	X ₂₄	Degree Centralization					✓		✓		✓	

Notes: The table shows the set of input variables included in each model, divided into control (in-field) and network-related variables. BLR represents the reduced logistic regression model, NB stands for the Naive Bayes approach, computed on the full input data and the same set of regressors included in the reduced logistic model (NB-RED); ANN gives the composition of the proposed artificial neural network input setting, and RF that of the random forest. All models were compared with their respective versions excluding the network indicators in the input models (-N).

a rich range of tactical solutions (average neighborhood). Our findings based on statistical models could benefit the overall team performance and can provide crucial guidelines supporting, in a practical way, human prediction and decision-making, especially concerning individual and team skills training.

Further developments of this approach should consider spatial information on the pitch, qualitative attributes of the connection, such as long versus short passes, and measures related to the network representing interactions among players of the two competing teams and measures related to the evolution of the passing structure during the match. This could also provide more information about the quality of defense actions and the overall match-level performances.

Finally, complexity and non-linearity of the passing relationship emerged in a very recent work [48]. In this sense, even if a machine learning approach may be suitable for complex and non-linear problems, it requires improvements that add data features, such as temporal information. Some machine learning methods, used in the field of Reproducing Kernel Hilbert space [49,50],

should be applied in the perspective of functional data analysis. These methods can be also updated through the use of more recent developments [51–53] and can find practical applications in analysis of sport data.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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